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Quantitative Methods for Humanities

Research Methods

2024/2025

R Requirements

Take the swirl lessons CAUSALITY1 and CAUSALITY2 from the swirl course `qss-swirl`

Introduction to Research Methods

What are research methods?

Research Method

Systematic process of inquiry that aims to discover, interpret, or revise facts, events, behaviors, or theories.

Empirical research

Empirical research

Research method that involves **collecting**, analyzing, and interpreting **data** to increase knowledge and understanding of a particular topic or phenomenon.

> It is a capital mistake to theorize before one has data.
>
> - Arthur Conan Doyle (writing as Sherlock Holmes)

Empirical research by type of data

- **Quantitative research:**
  - **Data type:** numerical, structured data.
  - **Analysis tools:** statistical and computational methods.
  - **Scope:** answers questions with measurable, objective data.
  - **Example:** studying the effect of minimum wage on unemployment rates.
  - **Pros:** precise, generalizable, and efficient for large datasets.
  - **Cons:** may oversimplify complex issues and lacks contextual depth.
- **Qualitative research:**
  - **Data type:** non-numerical, unstructured data (e.g., interviews, text).
  - **Analysis tools:** subjective interpretation.
  - **Scope:** explores deeper meanings and social phenomena.
  - **Example:** understanding voter perceptions of healthcare policies.
  - **Pros:** rich, detailed insights; flexible for new topics.
  - **Cons:** subjective, not easily generalizable, and time-intensive.

Quantitative research

Quantitative research can be divided in two main categories, in terms of how much control the researchers have over the **assigning mechanism** used to create the treatment and control group.

- **Experimental research:** the researcher has control over the assigning mechanism. Often uses primary data.
- **Observational research:** the researcher does not have control over the assigning mechanism. Often relies on secondary data

Validity of a research

The **validity of a research** refers to the **accuracy and generalizability** of its findings

- **External validity:** extent to which the results can be generalized to other settings, populations, or times.
  Question: Can the findings be applied outside the specific conditions of the study?
- **Internal validity:** extent to which the observed effects are due to the treatment variable and not to other external factors.
  Question: Did the study design and execution allow us to confidently say that variable T caused variable Y?

Balancing Internal and External Validity

The validity trade-off

- **High Internal Validity:** Typically achieved in tightly controlled experiments, often does not reassemble general scenarios and sacrifices external validity.
- **High External Validity:** Often achieved in real-world settings, but such studies might sacrifice control over variables and, consequently, may compromise internal validity.

Experimental Research

Experimental research

Experimental research

Quantitative research that examines how a treatment (independent variable) causally affects an outcome (dependent variable) by assigning varying values of the treatment variable to different observations, and measuring their corresponding values of the outcome variable.

Randomized control trials

Randomized Control Trials (RCTs)

Experimental research where each unit is randomly assigned either to the treatment or control group.

- **Advantage:** improved internal validity.
- **Weakness:** potential lack of external validity, impossibility to set up in many cases.

A case study. Racial discrimination in the job market

To illustrate **causal inference** within RCTs, we recall the "racial discrimination in the job market" question we asked ourselves in the previous chapter:

- Does racial discrimination exist in the U.S. labor market?
- Is this a discrimination due to race perse, or should it be attributed to other factors such as racial gaps in educational attainment?

It turns out that this study[1] was carried out in a brilliant manner that allowed randomization.

Case Study! ADD URL

[1] Marianne Bertrand and Sendhil Mullainathan (2004) Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination.

Avoiding confounding and selection bias through randomization

Question: In the study about the racial discrimination in the US job market, would you say that there is confounding and/or selection bias?

The power of randomization

The **randomization** of treatment assignment in RCTs **guarantees that the average difference in outcome between the treatment and control groups can be attributed solely to the treatment**, because the two groups are on average identical to each other in all pretreatment characteristics.

- The authors of the study about the racial discrimination in the US job market carried out a RCT by creating fictitious identical résumés.
- They also managed to manipulate immutable characteristics (raced) by relying on perceived characteristics (perceived race)

Estimating the SATE under RCTs

Under RCTs:

- Use the observed control group outcome as an estimate for the treatment group's counterfactual, which implies that

$$\overline{Y(0)} \approx \overline{Y}_C.$$

- Use the observed treatment group outcome as an estimate for the control group's counterfactual, which implies that

$$\overline{Y(1)} \approx \overline{Y}_T.$$

$\overline{Y}_T$ and $\overline{Y}_C$ represent the average outcome of the treatment and control group, respectively, that is,

$$\overline{Y}_T = \frac{1}{n_T} \sum_{i \in S_T} Y_i(1), \quad \overline{Y}_C = \frac{1}{n_C} \sum_{i \in S_C} Y_i(0).$$

Difference-in-means estimator

Difference-in-means estimator

The Difference-in-Means estimator $\widehat{\mathrm{SATE}}_{DiM}$ is defined as:

$$\widehat{\mathrm{SATE}}_{\mathrm{DiM}} = \overline{Y}_T - \overline{Y}_C.$$

DiM estimator

- **Advantages:**
  - **Simple & Transparent:** Easy to calculate and interpret.
  - **Effective in RCTs:** Unbiased when the control and treatment groups have balanced confounders (like in RCTs).
- **Weaknesses:**
  - **Confounding Bias:** Biased in non-randomized studies.
  - **No Covariate Adjustment:** Ignores other influencing variables.

Observational research

Observational studies

Observational Studies

Studies in which researchers do not conduct an intervention, but instead they simply observe naturally occurring events and collect and analyze the data.

- **Advantage:** more feasible than experimental studies, improved external validity.
- **Weakness:** potential lack of internal validity, impossibility of ruling out selection bias in many cases.

Quasi-experimental Studies

Observational studies whose assignment mechanism to treatment and control groups can be leveraged by researchers to approximate experimental conditions.

Example: Estimating the causal effect of additional schooling on earnings.

- **Potential Confounders:** e.g., more dedicated students may stay in school longer and perform better in labor market.
- **Challenge:** Randomized experiments are unethical as education decisions cannot be forced.
- **Quasi-Experiment Design:** Exploit changes in compulsory schooling laws.
- **Scenario:** A 1955 reform increases mandatory schooling from 8 to 10 years.
- **Solution to Confounders:** Students affected solely by the new law provide potential unbiased data for causal inference.

A case study. Minimum wage and employment

To illustrate **causal inference** within Observational Studies, we ask ourselves the following question[1]:

- Does increasing the minimum wage reduce employment as economic theory predicts?

Case Study! ADD URL

[1] David Card and Alan Krueger (1994) Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania. American Economic Review

Reducing the confounding bias through statistical control

Statistical control

Statistical control refers to the use of statistical techniques to account for the influence of confounding factors that could affect the relationship between the treatment and the outcome and, hence, mitigate the bias in the estimation of causal effects.

Subclassification

Subclassification

Statistical technique used to control for potential confounding factors by dividing a dataset into smaller, more homogeneous groups (subclasses) based on the levels or values of a pretreatment variable.

- **Advantage:** Simple and intuitive method.
- **Limitations:**
  - **Residual Confounding:** If subclasses are too broad, differences within a subclass may still bias results.
  - **Loss of Precision:** Dividing data into subclasses can reduce sample size within each group, lowering statistical power.
  - **Too many Interactions:** Considering all possible interaction may be too expensive.

Before-and-after design

Before-and-After (BA) design

A research method used to examine how the outcome variable changed from the pretreatment period to the posttreatment period for the same set of units.

Assumption (No Time-Varying Confounders): The BA design rests upon the assumption that there are not time-varying confounder factors.

Before-and-after design. Advantages and Weaknesses

- **Advantage:**
  - Allows to adjust for any confounding factor that is specific to each unit but does not change over time.
  - Needs relatively small data since the control group is taken to be the pretreatment treatment group.
- **Weakness:** does not address possible bias due to time-varying confounders.

Before-and-after design. Advantages and Weaknesses

Example: A study evaluates the impact of a new educational program on student performance:

- **Data:** Students' test scores are measured before and after the program is introduced.
- **Causal effect estimate:** The difference in scores indicates the program's effectiveness.
- **Assumption of No Time-Varying Confounders:**
  - **Met:** Curriculum, teacher quality, and/or testing conditions remain constant.
  - **Not Met:** A new teaching assistant joins, or students participate in a parallel tutoring program.

Before-and-after estimator

Before-and-After (BA) estimator

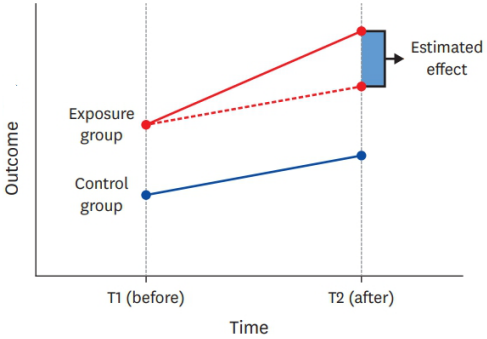
The BA estimator of the SATE, denoted by $\widehat{\mathrm{SATE}}_{BA}$ is computed as the $\widehat{\mathrm{SATE}}_{DiM}$, but by **identifying the pretreatment treatment group as the control group.**

Note that the BA design and estimators requires **measures before and after the treatment**. This type of data taken in different points in time is called **panel data**.

Difference-in-Differences (DiD) design

The DiD design extends the before-and-after design to address the confounding bias due to time trends.

Assumption (Parallel Trend): The key assumption behind the DiD design is that the outcome variable, both in the treatment and control group, follows a parallel trend in the absence of treatment.



DiD estimator. Advantages are weaknesses

- **Advantage:** Accounts for external time-varying confounders, as long as they affect treatment and control groups equally over time.
- **Weakness:** Biased estimation if the time-dependency of the confounders differs between the control and treatment group.

DiD estimator. An Example

Example: A study evaluates the impact of a new transportation subsidy on student attendance:

- **Data:** Attendance rates are measured for both students in districts with and without the subsidy before and after its introduction.
- **Causal effect estimate:** Before-and-after estimate minus the difference in the outcome for control group after and before the treatment
- **Assumption of Parallel Trends:**
  - **Met:** Factors like economic conditions, weather, or school policies influence attendance similarly in both districts over time.
  - **Not Met:** If one district experiences an unrelated policy change (e.g., stricter truancy enforcement) or a unique event (e.g., a school closure due to weather).

DiD estimator

The DiD estimator is defined as

$$\widehat{\mathrm{SATE}}_{DiD} = \underbrace{(\overline{Y}_T^{\text{after}} - \overline{Y}_T^{\text{before}})}_{\substack{\text{difference for the treatment group} \\ = \widehat{SATE}_{BA}}} - \underbrace{(\overline{Y}_C^{\text{after}} - \overline{Y}_C^{\text{before}})}_{\text{difference for the control group}},$$

where $\overline{Y}_T^{\text{after}}$ and $\overline{Y}_T^{\text{before}}$ refers to the pre and posttreatment sample average of the outcome in the treatment group, while $\overline{Y}_C^{\text{after}}$ and $\overline{Y}_C^{\text{before}}$ hold analogous definitions for the control group.